

Credit: NEANY



The Dragon Spy

floor mapping. The Turkish navy has developed the Ulaq AUSV which is deployed from a ship and designed to work with other UAVs, USVs, aircraft and warships. Armed with missiles and torpedos, it can also be configured as a firefighting ship. The AUSV is designed for ISR, armed escort, force protection, asymmetric warfare and surface warfare. India too is fishing around for developing an USV, with a focus on a mine counter-measures vehicle. The Indian Navy wants a versatile platform, capable of being operated in both manned and unmanned configurations. This USV is designed for being launched by a vessel at sea or from a harbour. The Venus USV deployed by the Singaporean coast guard flags and hounds suspicious vessels till the high speed interceptors catch up with them. The modular design allows for multiple payloads, including weapons.

The best example to demonstrate the future of USVs is American Sea Hunter, which is also known as the Anti-Submarine Warfare Continuous Trail Unmanned Vehicle (ACTUV) while in development by DARPA. This 40 meter long USV has two screws (propellers) and is designed to patrol the oceans for months at a time, with no crew on board. While the innards have space enough for repair and maintenance personnel to carry out their work, there are not enough facilities or space on board to maintain a full time crew at all. The Sea Hunter has a trimaran design, with two outrigger hulls on either

side of the main hull. The craft uses a combination of optical guidance and radar to avoid collisions with other vessels in the sea. Eventually the strategic vision is to populate the oceans with swarms of similar vessels, that remove the requirement of crewed warships entirely. Apart from ISR missions, the initial tests have already tested offensive anti submarine payloads, where the Sea Hunter can detect and engage with enemy submarines autonomously, or follow the submarines for months at a time. For this reason the Sea Hunter is also known as the submarine killer.

The Chinese seems to have constructed their own version of the Sea Hunter, which is slightly longer based on image analysis of photos and satellite imagery, as well as slightly narrower as the outrigger hulls are placed closer to the primary hull. At this point the capabilities of this craft is unclear, and it is also unknown if the vehicle was privately developed or constructed by the Chinese Military. The Chinese military are actively developing a range of USVs with offensive capabilities though, including the inflatable Tianxing-1 equipped with a machine gun, the Jari Catamaran that is equipped with anti submarine torpedos, a 30 mm cannon and surface

to air missiles and the Jing Hai USVs that are “research vehicles” which supported the Chinese Icebreaker Xuelong in antarctic missions with seafloor mapping, but with armed variants showcased at defence expos.

When it comes to regulating systems and platforms such as the Sea Hunter, one of the points of confusion is whether they should be categorised as weapons, or vessels, or something else entirely. Under international treaties, vessels that engage with an enemy are required to give quarter to an opponent who signals for surrender, rescue crew from shipwrecks, and search for missing persons reported by the enemy. Medical ships, science vessels and the ships of neutral countries are also protected by international treaties. It is not likely that USVs can fulfill all the requirements for vessels, as they are simply not designed to do so, but if they are treated as weapons then these same obligations do not apply, yet. Even when classified as a weapon, there is



Credit: DARPA

The Sea Hunter

a requirement to distinguish enemies from civilians, and use force proportional to the requirements. While patrolling, the USVs will be required to respect international navigation laws. The conventional laws and international treaties in place deal with either weapons or vessels, and as such do not cover the full capabilities of USVs in development, many of which are done in secret away from the view of the general public. **d**